

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-20-1 AC GEN 1 and AC GEN 2 VOLT Indications**
(on ESID)

Interval	Installed	Required	Procedure
C	6	0	NIL

May be inoperative provided:

- Associated electrical cautions are operative,
- Associated AC GEN LOAD Indication(s) are operative, and
- Both DC Generators are operative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative AC VOLT indication on ESID Electrical Page must be placarded in the flight compartment.

Operating Procedures

None required.

Maintenance Procedures

- Placard inoperative AC VOLT indication INOP on ESID Control Panel.
- Make appropriate entry in the aircraft journey log.

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Signed by: Manoj
Kumar Bajpai
Director
(Airworthiness)
Date: 07-Jun-
2024 13:52:49

24-20-1. 1

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-20-2 AC GEN 1 and AC GEN 2 LOAD Indications
(on ESID)**

Interval	Installed	Required	Procedure
C	6	0	NIL

May be inoperative provided:

- Associated electrical cautions and/or warnings are operative,
- Associated AC GEN VOLT Indications are operative, and
- Both DC Generators are operative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative AC VOLT indication on ESID Electrical Page must be placarded in the flight compartment.

Operating Procedures

None required.

Maintenance Procedures

- Placard inoperative AC LOAD INOP on ESID Control Panel.
- Make appropriate entry in the aircraft journey log.

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**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-20-3 #1 AC GEN and #2 AC GEN Caution Lights**

Interval	Installed	Required	Procedure
C	2	1	NIL

May be inoperative provided:

- Associated AC GEN VOLT and AC GEN LOAD Indications are operative and are periodically monitored during flight, and
- Both AC Generators are verified operative prior to each flight.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative "AC GEN" Fail Caution Lights must be placarded in the flight compartment.

Operating Procedures

None required.

Maintenance Procedures

- Placard "AC GEN" Fail Caution Light on CAUTION LIGHT panel.
- Make appropriate entry in the aircraft journey log.

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**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-1 Transformer Rectifier Units (TRUs)**

Interval	Installed	Required	Procedure
B	2	1	(M)

May be inoperative provided:

- Both DC Starter/Generators are operative, and
- The inoperative TRU is deactivated.

NOTE: The TRU is also considered inoperative if the TRU cooling fan is degraded, malfunctioning or inoperative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative Transformer Rectifier Unit (TRU) must be placarded in the flight compartment.

Operating Procedures

None required.

Maintenance Procedures

NOTE: If Step 3, in regards to disconnecting the TRU Fan is to be carried out, Approved Maintenance Personnel as defined by (M#) is required. Step 3 is a conditional statement and such does not modify the rest of the maintenance procedures to explicitly require maintenance personnel.

- Placard inoperative Transformer Rectifier Unit on the CAUTION LIGHT panel.
- Open and clip failed L or R TRU circuit breaker on the 115 VAC circuit breaker panel.
- If the TRU cooling fan is the root cause of the TRU being considered inoperative, or needs to be deactivated with the TRU, it must be disconnected from the DC Secondary Bus using the following steps:
 - Disconnect the impacted TRU from the DC Contactor Box at the terminal on the DC Contactor Box
 - Terminal 3 (T3) for left TRU connection to DC Contactor Box, or
 - Terminal 4 (T4) for right TRU connection to DC Contactor Box.

(...Continued)

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**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-1 Transformer Rectifier Units (TRUs)**

- b) Apply a non-conductive covering on both the disconnected electrical connections (Terminal on DC Contactor Box and disconnected wire from Terminal).
- c) Stow loose wire in a manner such that it cannot inadvertently be knocked loose and make contact with any other connections within its vicinity.
4. Start Engines 1 and 2 and select CONDITION LEVERS above START/FX.
5. At DC CONTROL panel select the following to OFF POSITION:
 - a) GEN 1
 - b) GEN 2
 - c) EXTERNAL POWER
6. Ensure that battery switches are selected to BATTERY MASTER, MAIN BATT, AUX BATT, STBY BATT.
7. At AC CONTROL panel select GEN 1 and GEN 2 to OFF.
8. Check the following caution lights are on:
 - a) # 1 DC GEN
 - b) # 2 DC GEN
 - c) # 1 AC GEN
 - d) # 2 AC GEN
 - e) L TRU
 - f) R TRU
9. Ensure MAIN BATTERY, AUX BATTERY, STBY BATTERY caution lights are off.
10. At DC CONTROL panel select GEN 2 switch to GEN 2. Check that # 2 DC GEN caution light goes off.
11. At ESID Electrical page check that L and R MAIN and L and R ESS bus voltage reads 28.0 ± 2 Volts, and L SEC and R SEC bus voltage reads zero and the MAIN BATT, AUX BATT, STBY BATT LOAD indicates a charging current.
12. At ESID Electrical page, ESID LOAD indicator for DC GEN 2 displays load demand.
13. At DC CONTROL panel select GEN 1 switch to GEN 1. Check the # 1 DC GEN caution light goes off.
14. At ESID DC Electrical page check that all bus voltages read 28.0 ± 2 Volts.
15. At ESID Electrical page, LOAD indicator for DC GEN 1 displays load demand.
16. At AC CONTROL panel select GEN 1 or GEN 2 on.
17. Check appropriate L or R TRU caution light for operative TRU goes off.
18. At ESID Electrical monitor page, check that L SEC and R SEC bus reads 28 ± 2 Volts.
19. At ESID Electrical page, check LOAD indicator displays load demand of operative L TRU or R TRU.
20. Shutdown the engines.
21. Select BATTERY MASTER, MAIN BATT, AUX BATT and STBY BATT switches OFF.
22. Make appropriate entry in the aircraft journey log.

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24-30-1. 2

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-2 DC Generator Control Unit - Start / Termination Function**

Interval	Installed	Required	Procedure
B	2	0	(O)

May be inoperative provided the start is manually terminated at 50% NH.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative DC Generator control Unit must be placarded in the flight compartment.

Operating Procedures

1. Carry out associated engine(s) start in sequence in accordance with AFM Section 4 – NORMAL AND ABNORMAL PROCEDURES, item 4.1.5 – ENGINE START PROCEDURE.
2. Closely monitor associated NH indicator during starting and terminate start function by deselecting ENGINE START SELECT switch to center SELECT position (off) when NH indication shows positive acceleration beyond 50%.

Maintenance Procedures

1. Placard Engine Start SELECT switch on Engine Start panel on overhead console.
2. Make appropriate entry in the aircraft journey log.

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**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-3 BATT °C Temperature Indications**

Interval	Installed	Required	Procedure
C	3	0	(O)

May be inoperative provided:

- The associated BAT HOT Warning Light(s) are verified operative, and
- The associated DC BATT LOAD Indication(s) are verified operative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative Battery Temperature Indicator(s) must be placarded in the flight compartment.

Operating Procedures

- Perform test of Caution and Warning Panel to check that BAT HOT warning light illuminates.
- Respective battery load indications are monitored for any trend increasing charging current (+).
- If a battery overheat condition occurs, land immediately at the nearest suitable airport.

Maintenance Procedures

- Placard ESID Systems Display with "INOP" in the flight compartment.
- Make appropriate entry in the aircraft journey log.

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24-30-3. 1

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-4 APU Generation System****24-30-4-1**

Interval	Installed	Required	Procedure
D	1	0	(M#)

May be inoperative provided:

- The cause of the malfunction is determined, and
- Appropriate action is taken to ensure that no hazard exists.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative APU Generation System must be placarded on the APU Control Panel near the GEN button.

Operating Procedures

None required.

Maintenance Procedures

- At the tailcone, open the APU compartment access doors and visually confirm integrity of electrical, fuel, pneumatic connections, and fire detection loop.
- Close APU access door.
- Perform fire detection test of APU loop.
- Start and run APU, confirm the "APU fault light" is not illuminated.
- At the APU Control Panel, when the PWR "RUN" light is illuminated, select the "BLEED" switch to on and confirm airflow from vents.
- Deselect "BLEED" and shut down APU.
- Placard the APU GEN button on the APU Control Panel.
- Make appropriate entry in the aircraft journey log.

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Director
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24-30-4. 1

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-4 APU Generation System****24-30-4-2**

Interval	Installed	Required	Procedure
D	1	0	NIL

May be inoperative provided the APU is considered inoperative and is not used.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative APU Generation System must be placarded on the APU Control Panel near the GEN button.

Operating Procedures

None required.

Maintenance Procedures

1. Placard the APU GEN button on the APU Control Panel.
2. Make appropriate entry in the aircraft journey log.

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Kumar Bajpai
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24-30-4. 2

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-5 DC GEN LOAD Indications****24-30-5-1**

Interval	Installed	Required	Procedure
C	2	1	(O)

May be inoperative provided both DC Generators are operative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative DC GEN Load indicator must be placarded in the flight compartment.

Operating Procedures

Ensure all DC generators and TRUs operate normally using the following steps:

1. Check the load indications for the remaining channels.
2. Observe correct annunciation of caution lights, ie. All DC GEN and TRU caution lights out.

Maintenance Procedures

1. Placard DC GEN LOAD Indication on the ESID.
2. Make appropriate entry in the aircraft journey log.

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Director
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**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-5 DC GEN LOAD Indications****24-30-5-2**

Interval	Installed	Required	Procedure
C	2	1	(O)

May be inoperative provided the inoperative Indication is associated with a failed DC Generator.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative DC GEN Load indicator must be placarded in the flight compartment.

Operating Procedures

Ensure all DC generators and TRUs operate normally using the following steps:

1. Check the load indications for the remaining channels.
2. Observe correct annunciation of caution lights, i.e., All DC GEN and TRU caution lights out.

Maintenance Procedures

1. Placard DC GEN LOAD Indication on the ESID.
2. Make appropriate entry in the aircraft journey log.

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24-30-5. 2

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-30-6 DC Starter/Generation Systems**

Interval	Installed	Required	Procedure
B	2	1	(M#)(O)

One GCU and/or Generator may be inoperative in the generation mode only (i.e. DC GEN caution light illuminated) provided:

- The cause of the malfunctions is determined and appropriate action is taken to ensure that no hazard exists,
- The failed DC Generator is switched OFF, and
- Both Transformer Rectifier Units (TRUs) are verified operative, and
- Operations are conducted in compliance with AFM Supplement 92, OPERATION WITH ONE DC GENERATOR INOPERATIVE.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative DC Starter/Generator (generation mode inoperative) must be placarded at the DC CONTROL panel in the flight compartment.

Operating Procedures

- Verify both transformer rectifier units (TRUs) are operating normally. Ensure L TRU and R TRU caution lights are out.
- Ensure Battery Master, Main Battery, Auxiliary Battery and Standby Battery switches are selected to BATTERY MASTER, MAIN BATT, AUX BATT and STBY BATT positions respectively on the DC CONTROL panel.
- Operations are to be conducted in compliance with AFM Supplement 92, OPERATION WITH ONE DC GENERATOR INOPERATIVE.

Maintenance Procedures

- Perform a visual inspection of inoperative starter/generator and surround for signs of overheat, leakage, wire damage and general conditions.
- Start engine on affected side (i.e. side with DC Generator inoperative) to verify the start function operates normally and select power lever to FLT IDLE. Ensure all AC and DC electrical system circuit breakers are closed.

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24-30-6. 1



DHC-8-402

MINIMUM EQUIPMENT LIST

ELECTRICAL POWER

ATA 24

24-30-6

DC Starter/Generation Systems

3. Ensure Battery Master, Main Battery, Auxiliary Battery and Standby Battery switches are selected to BATTERY MASTER, MAIN BATT, AUX BATT and STBY BATT positions respectively and select EXTERNAL POWER to OFF, on the DC CONTROL panel.
4. Select the DC GEN switch, failed generator, to OFF position on the DC CONTROL Panel.
5. At AC CONTROL panel select associated generator switch to GEN 1 or GEN 2 position.
6. Check that the respective AC GEN caution light goes out.
7. Select Electrical page on ESID. Check for 115 VOLTS indication on all phases.
8. Ensure TRU caution lights are out.
9. At ESID Electrical page, check that DC LOAD meter indicates a load when L TRU and R TRU are selected respectively and the L SEC and R SEC buses indicate 28 ± 3 volts on the DC BUS VOLTS indicator when selected to associated DC bus.
10. Shut down engine.
11. Placard inoperative DC Starter/Generator (generation mode inoperative) at the DC CONTROL panel in the flight compartment.
12. Make appropriate entry in the aircraft journey log.

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24-30-6. 2

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-40-1 AC External Power System**

Interval	Installed	Required	Procedure
D	1	0	NIL

May be inoperative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative AC External Power System must be placarded in the flight compartment.

Operating Procedures

None required.

Maintenance Procedures

1. Placard EXT POWER switch on AC CONTROL panel on overhead console.
2. Make appropriate entry in the aircraft journey log.

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Kumar Bajpai
Director
(Airworthiness)
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24-40-1. 1

**DHC-8-402****MINIMUM EQUIPMENT LIST*****ELECTRICAL POWER******ATA 24*****24-40-2 DC External Power System**

Interval	Installed	Required	Procedure
D	1	0	NIL

May be inoperative.

Minimum Equipment List Procedures (MELP)**Placard**

Inoperative DC External Power System must be placarded in the flight compartment.

Operating Procedures

None required.

Maintenance Procedures

1. Placard EXT POWER switch on DC CONTROL panel on overhead console.
2. Make appropriate entry in the aircraft journey log.

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24-40-2. 1